

HONG KANG

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EDUCATION

McGill University <i>Doctor of Philosophy</i> • Major: Computer Science • Overall GPA: 4.00/4.00	Jan. 2025 - Now
The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) <i>Master of Philosophy</i> • Major: Computer and Information Engineering • Overall GPA: 3.78/4.00	Sep. 2021- Nov. 2023
University of Glasgow <i>Bachelor of Engineering with Honors of the First Class</i> • Major: Electronics and Electrical Engineering • Overall GPA: 17.31/22.00	Sep. 2017-Jul. 2021
University of Electronic Science and Technology of China (UESTC) <i>Bachelor of Engineering</i> • Major: Electronic Information Engineering, Department: Glasgow College • Overall GPA: 3.53/4.00, Junior-Senior GPA: 3.76/4.00 • Honors: Outstanding Student Leader Scholarship, UESTC, 2020 Model Student Scholarship, UESTC, 2019, 2020	Sep. 2017-Jul. 2021

AECADEMIC PAPER

- Zhao Rong, **Kang Hong**, Zhang Qiyue, Fan Sizheng and Cai Wei. "LocPoS: A Location-Based Proof-of-Stake Mechanism for Industrial Digital Twins." *IEEE Transactions on Computational Social Systems*, 2025.
- Zhao Rong, Liu Yunshu, **Kang Hong**, and Cai Wei. "A Transaction Fee Mechanism for Resource Allocation in Blockchain-Based IoT Systems". *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2025.
- Lin Lehao, **Kang Hong**, Shi Yuqi, Duan Haihan, Abdulmotaleb El Saddik, and Cai Wei, "ASimp: Automatic High-Poly 3D Mesh Simplification for Preprocessing Based on QoE", In *IEEE International Conference on Multimedia & Expo (ICME'25)*, Nantes, France, June 30 - July 4, 2025.
- Sun Jinghan, Long Hou-Wan, **Kang Hong**, Fang Zhixuan, Abdulmotaleb El Saddik and Cai Wei, "Contract Design for SCaaS with Multi-dimensional Private Information", *IEEE Transactions on Computational Social Systems*, 2025.
- **Kang Hong**, Li Minghao, Lin Lehao, Fan Sizheng, and Cai Wei, "Bridging Incentives and Dependencies: An Iterative Combinatorial Auction Approach to Dependency-Aware Offloading in Mobile Edge Computing", *IEEE Transactions on Mobile Computing*, 2024.
- Lin Lehao*, **Kang Hong***, Sun Xinyao, and Cai Wei, "SemNFT: A Semantically Enhanced Decentralized Middleware for Digital Asset Immortality", In *32nd ACM International Conference on Multimedia (MM'24)*, Melbourne, Australia, October 28-November 1, 2024.
- **Kang Hong**, Li Minghao, Fan Sizheng, and Cai Wei, "Combinatorial Auction-enabled Dependency-Aware Offloading Strategy in Mobile Edge Computing", *IEEE Wireless Communications and Networking Conference (WCNC 2023)*.
- **Kang Hong**, Zhang Zaixin, Dong Junyi, Xu Hao, Valente Klaine Paulo, and Zhang Lei "Blockchain-enabled COVID-19 Contact Tracing Solutions", *Wireless Blockchain: Principles, Technologies and Applications*, Wiley-IEEE Press.
- **Kang Hong***, Zhang Zaixin*, Dong Junyi, Ji Yinghao, Xu Hao, and Zhang Lei. "BeepTrace for COVID-19 Pandemic: a Demo.", *3rd Conference on Blockchain Research & Applications for Innovative Networks and Services (BRAINS 2021)*

RESEARCH EXPERIENCE

Explainable Airdrop Hunter Detection (Working On)

Instructor: Prof. Xue Liu, McGill University

- Developed a multi-level contrastive learning framework (node-level, subgraph-level, and class-level) to enhance representation quality and detection robustness;
- Building an open-source dataset based on Hop Protocol airdrop event;
- Plan to design an explainer to help the community better understand the account features and transaction relationships of airdrop hunters. The goal is to provide a trustworthy and interpretable method, while also introducing false positive cases as much as possible to reduce user complaints.

Role-Aware Reinforcement Learning for Strategic Play in Social Deduction Games

Course Project, McGill University

- Designed a modular Judger–Chooser system combining reinforcement learning (REINFORCE, PPO, LinUCB) with frozen LLM outputs to play the “Go Undercover” social deduction game;
- Built a phase-aware reward model (Judger) scoring LLM responses based on strategic metrics like deception, persuasion, and information gain;
- Developed and evaluated Chooser policies using gameplay logs, enabling role- and phase-conditioned utterance selection under partial observability;
- Demonstrated that REINFORCE achieved the best balance of Judge score and win rate in real-game simulations, while LinUCB offered fastest convergence;
- Highlighted asymmetry in role difficulty and uncovered a gap between maximizing response quality and achieving long-term strategic wins.

Benchmarking Preference Alignment Algorithms Under Differential Privacy Constraints

Course Project, McGill University

- Investigated how differential privacy (DP) affects preference alignment in LLMs by benchmarking PPO-based and one-stage RLHF methods (e.g., DPO, etc) under identical privacy budgets;
- Implemented differentially private PPO pipelines using DPSGD and parallel composition theory to control cumulative privacy loss across training stages;
- Contributed to reward model and policy fine-tuning under DP constraints, using the Anthropic HH-RLHF dataset and LoRA for efficient adaptation;
- Demonstrated that one-stage algorithms achieve higher alignment scores than PPO under the same DP budget due to more efficient data reuse;
- Analyzed the performance impact of reference-free optimization and response length normalization, guiding privacy-preserving LLM alignment design.

Building a Decentralized Framework Addressing NFT Storage and Verification Challenge

Instructor: Prof. Wei Cai, The Chinese University of Hong Kong, Shenzhen

- Investigated the NFT market and identified storage inefficiencies, with only 9% of NFTs stored on-chain and the rest risk asset linkage losses on private servers or IPFS;
- Proposed a decentralized NFT storage and verification framework leveraging deep learning (autoencoders) and blockchain, aiming for efficient storage and reduced dependency on IPFS/private servers;
- Designed and implemented the framework to be compatible with existing NFT standards like ERC721, ensuring seamless integration into the current ecosystem;
- Validated the proposed framework through performance evaluations, showcasing its superiority in terms of reconstruction, storage costs, and integration capabilities.

Mechanism Design for Computation Partitioning and Offloading in Mobile Edge Computing (MEC)

Instructor: Prof. Wei Cai, The Chinese University of Hong Kong, Shenzhen

- Recognized the absence of an incentive mechanism in current offloading methods for dependent tasks in MEC and highlighted the potential of integrating auction theory as a solution;
- Proposed a novel multi-stage iterative combinatorial double auction (MICDA) mechanism. This integrates dependency-aware task offloading with combinatorial auction dynamics under the Cloud-Edge-End cooperative computing paradigm;
- The MICDA mechanism, underpinned by a concurrent provider and consumer (CPC) model, employs a two-stage approach:

- Stage 1: Utilizes a winner determination algorithm, informed by market clearing bid densities, to determine auction winners and compute payments/revenues for mobile devices and service providers, respectively;
- Stage 2: Applies linear relaxation with the ellipsoid method, employing a hypergraph-based proxy cost model for efficient demand oracle queries. This ensures polynomial time complexity for the winner determination problem. Post-processing utilizes the Lavi-Swamy decomposition technique and adopt the Vickrey–Clarke–Groves (VCG) payment rule to achieve a truthful-in-expectation mechanism;
- Demonstrated through theoretical analysis that the MICDA mechanism satisfies essential economic properties like truthfulness, individual rationality, weak budget balance, and polynomial time complexity;
- Experimental evaluations indicate MICDA's superiority against non-dependency-aware offloading strategies and its competitive edge over other dependency-aware algorithms in performance metrics.

Decentralized Application Development for COVID-19 contact tracing

Instructor: Prof. Lei Zhang, University of Glasgow

- Develop a mobile APP for Covid-19 cases tracing while ensuring the users' privacy;
- Set up the Ethereum environment on several AWS distributed nodes and connect them to establish the private chain;
- Applied solidity to assist the users to upload and load Hash encrypted data information;
- Applied JAVA language and Ethereum Web3J API to realize the interaction process between users and blockchain remotely through Http port;
- Emerged block chain with the APP to complete the function of BeepTrace and verified the performance and handling capacity of the blockchain;
- Interviewed and broadcast by BBC.

Development of a Smart Contract for Route Recommendation Service on Ethereum by Solidity

Instructor: Prof. Lei Zhang, University of Glasgow

- Applied blockchains to write a smart contract for route recommendation Ethereum by Solidity;
- Established a private Ethereum chain locally (Ganache+Remix IDE) to verify the feasibility of the smart contract;
- Verified the smart contract could recommend the most time-efficient route for users to travel, meanwhile, it solves the problems of privacy leakage and data reliability of traditional third-party mapping application.

Smart Contract Development for Toshiba Europe

- Developed a smart contract in Solidity to interact data from sensors and scanners with the blockchain platform;
- Designed the structure of data storage within the contract to make the data in the block chain be accessed more clearly and efficiently;
- Realized the conversion between the timestamp and the regular time of Ethereum to make recorded time could be read directly.

Academic Services

- **Reviewer:** IEEE/ACM Transactions on Networking (ToN), IEEE Transactions on Cloud Computing (TCC), IEEE Transactions on Mobile Computing (TMC).

ADDITIONAL INFORMATION

- **Language Skills:** Chinese Mandarin (Native), English (IELTS: 7.0)
- **Computer Skills:** Python, MATLAB, Solidity, Latex, etc.